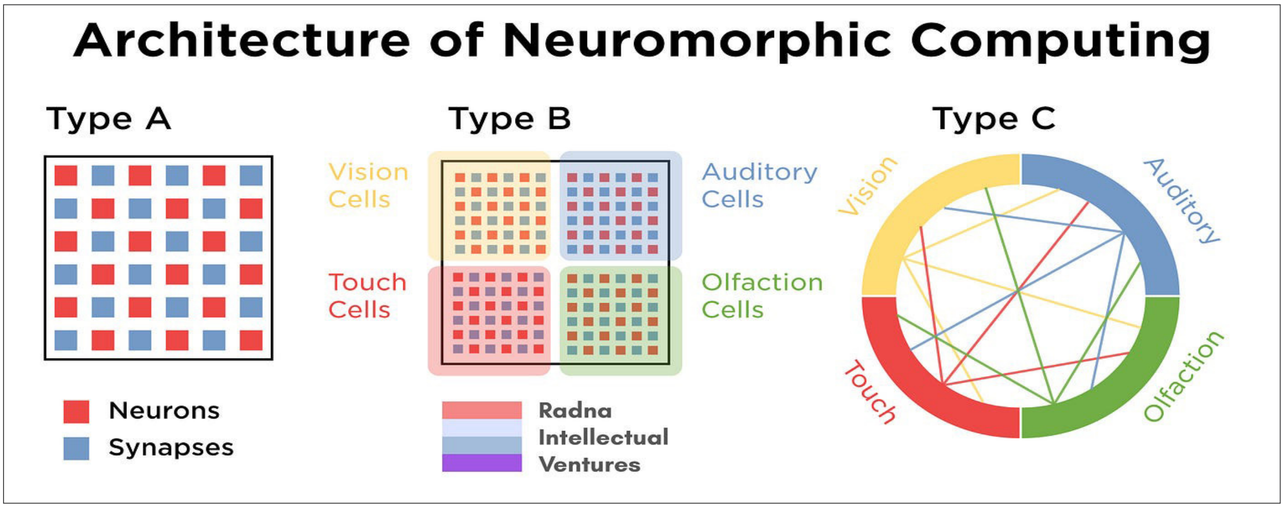

Figure 2: The architectural overview of neuromorphic computing (Google Images)

adapt on-the-fly, make real-time decisions at the edge without relying on centralised servers, and do so with remarkable power efficiency.

In essence, neuromorphic computing isn't just an incremental improvement over existing architectures; it represents a fundamental paradigm shift in how we approach computing for complex, interconnected systems like IoT.

Cross-comparing von Neumann and neuromorphic architectures

The architectural design behind any computing system not only defines its operational methodology but also its potential and limitations. So let us cross-compare the von Neumann architecture with the evolving architecture of neuromorphic computing. Figure 3 provides a cross-comparison between these architectures.

- ***von Neumann architecture***

Sequential processing: At its core, the von Neumann model is built around the fetch-decode-execute cycle. Instructions and data, both stored in memory, are fetched and processed sequentially, leading to inherent delays.

Shared memory bottleneck: The shared memory for both data and instructions often becomes a bottleneck, especially when dealing with complex computations. This is the infamous 'von Neumann bottleneck' that can limit system performance.

Stateful design: von Neumann systems operate in a stateful manner—operations depend on a sequence of previously executed instructions, leading to potential inefficiencies in certain real-time or parallel processing tasks.

- ***Neuromorphic architecture***

Parallelism at heart: Instead of a linear sequence, neuromorphic designs embrace massive parallelism, with artificial neurons operating concurrently. This mirrors the human brain, where billions of neurons can fire or stay silent simultaneously, allowing for multitasking and rapid sensory processing.

Event-driven modality: Neuromorphic systems, especially those using spiking neural networks (SNNs), operate based on events. A neuron either spikes (fires) or doesn't, making the system inherently efficient as operations are based on events rather than continuous cycles.

Inherent adaptability: By using components like memristors that can adjust their resistance based on historical currents, neuromorphic designs allow for an adaptability that's hardwired into the system. This is akin to how biological synapses can strengthen or weaken over time, allowing for learning and memory.

Why it matters for IoT

The transition from von Neumann to neuromorphic isn't just about mimicking the brain but about addressing real-world challenges. As IoT devices proliferate, the demands for low latency, real-time processing, and adaptive learning are increasing. The latency inherent in the von Neumann design, especially in large-scale data processing scenarios, can be a limiting factor for IoT devices needing real-time responses. Neuromorphic designs, with their parallelism and event-driven nature, can address these challenges, offering a more responsive and adaptable architecture for the next generation of IoT solutions.

Industry examples

As academia has unveiled the principles of neuromorphic computing, industry giants and innovative startups have taken the lead in actualising these concepts.